Evolutionary Design of Modular Soft Robot Joint

IEEE Publication Technology, *Staff, IEEE,*

***Abstract*—This document describes the most common article elements and how to use the IEEEtran class with LATEX to produce files that are suitable for submission to the IEEE. IEEEtran can produce conference, journal, and technical note (correspondence) papers with a suitable choice of class options.**

***Index Terms*—Article submission, IEEE, IEEEtran, journal, LATEX, paper, template, typesetting.**

I. Introduction

THIS file is intended to serve as a "sample article file" for IEEE journal papers produced under LATEX using IEEEtran.cls version 1.8b and later. The most common elements are covered in the simplified and updated instructions in "New_IEEEtran_how-to.pdf". For less common elements you can refer back to the original "IEEEtran_HOWTO.pdf". It is assumed that the reader has a basic working knowledge of LATEX. Those who are new to LATEX are encouraged to read Tobias Oetiker's "The Not So Short Introduction to LATEX," available at: http://tug.ctan.org/info/lshort/english/lshort.pdf which provides an overview of working with LATEX.

II. DATASET

A. Data Sources and Low-Annotation Protocol

In this study, we construct a low-annotation-budget polyp segmentation benchmark using multiple public colonoscopy datasets. The training universe is built from Kvasir-SEG, CVC-ClinicDB, and selected training centers of PolypGen. Kvasir-SEG provides 1000 polyp images with pixel-wise annotations, CVC-ClinicDB contains 612 annotated images, and PolypGen offers a large-scale multi-center resource with 1537 still images, 2225 positive video sequences, and 4275 negative frames.To further improve cross-center diversity,we use still images together with sparsely sampled positive video frames from the PolypGen. By combining these datasets, we aim to cover broader variations in lesion appearance, imaging conditions, and center-specific domain shifts, thereby providing a more realistic setting for evaluating label-efficient medical image segmentation.

Although several source datasets provide full pixel-wise annotations, we intentionally expose only a small subset of labeled samples during training in order to simulate the practical scenario of limited expert supervision. Specifically, the training universe is partitioned into a small labeled set L_{small} and a large unlabeled set U_{large}, where L_{small} is used for teacher adaptation and supervised student learning, while U_{large} is used for pseudo-label generation and semi-supervised expansion, as shown in Fig. 1 This protocol explicitly models the realistic clinical constraint that acquiring new expert masks is expensive and time-consuming, whereas unlabeled endoscopic images are comparatively abundant.

B. Ground-Truth Annotation and Pseudo-Label Construction

For samples in L_{small}, the ground-truth masks are taken from the officially released annotations or from expert-verified masks, depending on the source dataset and the final experimental setting. These annotations serve as the only human-supervised pixel-level references used to adapt the teacher model and anchor the student model to clinically reliable boundaries. In this way, the proposed framework does not assume abundant manual annotation, but instead treats a limited number of high-quality masks as the core supervision resource.

For the unlabeled set U_{large}, pseudo labels are generated by the domain-adapted MedSAM teacher. To improve pseudo-label reliability, a quality-control pipeline is introduced before these masks are used for student training. Specifically, low-quality predictions are filtered based on confidence, connected-component consistency, boundary smoothness, and region-size plausibility. High-confidence pseudo labels are directly included in the expanded training pool, medium-confidence samples are optionally corrected through rapid manual inspection, and low-confidence cases are discarded or reserved as hard samples for later refinement. This design forms a data flywheel in which a small amount of expert knowledge is progressively amplified into scalable supervision.

C. External Test Sets and Generalization Evaluation

To rigorously evaluate out-of-distribution performance, external testing is conducted on ETIS-LaribPolypDB, CVC-ColonDB, CVC-300, and held-out centers from PolypGen (C1, C2, C3, C4). ETIS-LaribPolypDB is used as a challenging external benchmark due to its difficult lesion appearances and substantial distribution gap from common training sets. CVC-ColonDB and CVC-300 serve as classical cross-dataset evaluation sets, while the held-out PolypGen centers C5 and C6 are used to assess cross-center generalization under multi-source domain shift. None of these external sets participate in model selection or hyperparameter tuning.

This evaluation protocol is designed to measure not only in-domain segmentation accuracy but also the robustness of the proposed framework under realistic deployment conditions. Since medical endoscopy systems often face differences in equipment, operators, centers, and imaging styles, strong external generalization is an essential criterion for assessing the practical value of the proposed low-annotation-budget framework.

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TABLE I
SUMMARY OF THE DATASETS USED IN THIS STUDY

Usage	Dataset	Training	Test	Total
Shared training universe	Kvasir-SEG	$L_{\text{small}}(10\%)$, $U_{\text{large}}(90\%)$	–	1000
	CVC-ClinicDB	$L_{\text{small}}(10\%)$, $U_{\text{large}}(90\%)$	–	612
	PolypGen (C1/C2/C3/C4)	U_{large} (100%)	–	1241
	Total	$L_{\text{small}} = 161$, $U_{\text{large}} = 2692$	–	2853
Strict external test	ETIS-LaribPolypDB	–	All	196
	CVC-ColonDB	–	All	380
	CVC-300	–	All	60
	PolypGen holdout (C5/C6)	–	All	296

L_{small} contains 161 expert-annotated images sampled only from Kvasir-SEG and CVC-ClinicDB.
U_{large} contains 2692 unlabeled images from Kvasir-SEG, CVC-ClinicDB, and PolypGen (C1–C4).
To prevent data leakage, images from the same patient, sequence, or held-out center were assigned exclusively to one subset.

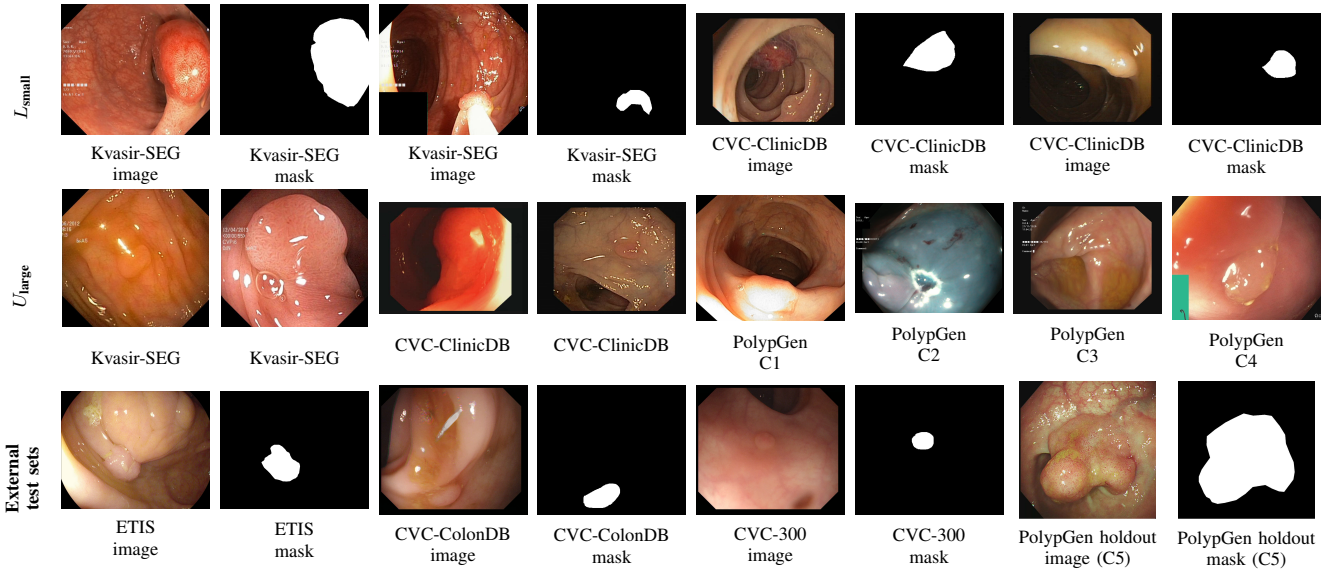

Fig. 1. Representative samples from the datasets used in this study. The first row shows annotated samples from the labeled subset L_{small}, which contains expert-verified image–mask pairs from Kvasir-SEG and CVC-ClinicDB. The second row shows representative unlabeled samples from U_{large}, including Kvasir-SEG, CVC-ClinicDB, and PolypGen training centers C1–C4. The third row shows representative external test samples from ETIS-LaribPolypDB, CVC-ColonDB, CVC-300, and PolypGen held-out centers (C5/C6), together with their corresponding ground-truth masks.

D. Dataset Splitting

To avoid information leakage, all samples are split at the patient level or sequence level whenever such metadata are available. Images or frames from the same patient, the same examination, or the same video sequence are assigned exclusively to a single subset. Within the training universe, L_{small} and U_{large} are constructed after this leakage-aware partitioning, ensuring that the teacher adaptation, pseudo-label generation, and student training are all performed under a consistent and reproducible protocol.

In practice, the exact proportion of L_{small} can be varied to simulate different annotation budgets, such as ultra-low-label, low-label, and moderate-label settings. This enables systematic analysis of how the proposed teacher-student framework behaves as the amount of real supervision changes, and whether the performance gain mainly comes from teacher adaptation, pseudo-label expansion, or student-side architectural improvements.

III. METHODOLOGY

A. Problem Definition

The goal of this study is to perform automatic polyp segmentation under limited annotation budget. Let $x \in \mathbb{R}^{H\times W\times 3}$ denote an input colonoscopy image and $y \in \{0,1\}^{H\times W}$ denote its binary segmentation mask. Under the low-annotation protocol, only a small labeled set $L_{small} = \{(x_i, y_i)\}_{i=1}^{N_l}$ is available, together with a much larger unlabeled set $U_{large} = \{x_j\}_{j=1}^{N_u}$, where $N_l \ll N_u$. The objective is to learn a deployable student model that achieves accurate and efficient segmentation by combining limited ground-truth supervision with teacher-generated pseudo supervision.

Unlike conventional fully supervised segmentation, our setting explicitly addresses the conflict between annotation scarcity and deployment efficiency. MedSAM provides strong medical segmentation priors but is computationally heavy and less suitable for fast clinical deployment, whereas lightweight task-specific networks can be efficiently deployed but usually require substantial annotated data. Therefore, the proposed method aims to transfer the teacher's prior knowledge into an

efficient student network through domain adaptation, pseudo-labeling, and knowledge distillation.

B. Overall Teacher-Student Framework

The overall framework follows a teacher-student paradigm with a closed-loop data expansion strategy. First, a MedSAM-based teacher is adapted to the polyp domain using a small number of expert-annotated samples. Second, the adapted teacher is used to generate pseudo labels for the large unlabeled pool. Third, a frequency-decoupled ResSwinUNet student is trained on both real labels and high-quality pseudo labels, optionally under additional distillation supervision from the teacher. In this way, the framework forms a complete pipeline from scarce annotation to scalable training and finally to efficient deployment.

The proposed framework decouples the roles of foundation-model supervision and task-specific deployment. The teacher model is used to absorb scarce expert knowledge and convert it into scalable pseudo supervision and soft structural priors, whereas the student model is designed to serve as the final inference engine. This design addresses the practical contradiction that MedSAM provides strong medical segmentation priors but is not ideal for real-time deployment due to its interaction cost and computational overhead

C. SAM-LoRA Teacher Adaptation and Pseudo-Label Generation

The teacher model is initialized from MedSAM and adapted to the target polyp domain using parameter-efficient low-rank adaptation (LoRA). Instead of updating all parameters, LoRA injects trainable low-rank matrices into selected attention-related layers, allowing the model to learn domain-specific appearance priors with substantially reduced optimization cost. In this study, the adapted teacher is expected to better capture polyp-specific color distribution, mucosal texture, shape variation, and blurry lesion boundaries than the original generic model.

During pseudo-label generation, box prompts, point prompts, or candidate regions from a coarse localization stage are used to trigger mask generation on unlabeled images. The teacher is therefore not treated merely as an annotation tool, but as a prior-rich supervisory model whose output can be filtered, refined, and subsequently transferred to the student. This role transformation is essential for converting scarce expert masks into scalable supervision.

D. Two-Round SAM-LoRA Data Flywheel

To expand supervision beyond L_{small}, we construct a two-round SAM-LoRA data flywheel. In Round 0, MedSAM is adapted on L_{small} to obtain the initial teacher T_0. In Round 1, T_0 is applied to the entire unlabeled pool U_{large}, and the resulting masks are filtered according to confidence, boundary quality, connected-component consistency, and region plausibility. The top 30%–40% high-quality pseudo-labeled samples are then added into the training pool. In Round 2, the remaining difficult unlabeled samples are reprocessed using the improved teacher or the intermediate student-assisted system, and an additional 10%–20% of reliable pseudo-labeled samples are incorporated. After the second round, the flywheel stops rather than expanding indefinitely.

In practical terms, the two-round design reflects a deliberate trade-off between supervision gain and noise accumulation. High-confidence pseudo labels are directly accepted, medium-confidence masks may undergo fast manual correction, and low-confidence predictions are moved into a hard-sample pool. The expansion process is terminated once the gain becomes saturated, when the external validation Dice improves by less than a predefined threshold, or when the manual spot-check pass rate of newly added pseudo labels falls below an acceptable level.

E. Frequency-Decoupled ResSwinUNet Student Network

As shown in Fig. 2, the proposed student network adopts a hybrid CNN-Transformer encoder-decoder architecture, termed ResSwinUNet, to balance local representation, global contextual modeling, and deployment efficiency. The overall framework consists of a shallow residual convolutional encoder, two Swin-based semantic stages, a wavelet bottleneck, and a U-shaped decoder with attention-gated skip connections. Instead of simply stacking standard modules, the design is tailored to polyp segmentation, where subtle lesion boundaries, mucosal texture interference, and cross-domain appearance variation must be handled simultaneously.

The encoder follows a progressive hybrid design. As illustrated in Fig. 2(a) and Fig. 2(b), given an input image $I \in \mathbb{R}^{H\times W\times 3}$, the shallow ResNet branch first extracts multi-scale convolutional features $\{x0, x1, x2\}$, which is responsible for extracting stable low-level visual patterns, including mucosal texture, specular reflections, and coarse lesion boundaries. And the subsequent Swin stages strengthen long-range semantic interaction at deeper levels, each Swin stage begins with a strided convolutional downsampling layer and is followed by a stack of Swin blocks to progressively refine semantic representations.This combination is motivated by the fact that polyp regions are often locally ambiguous but globally distinguishable only in relation to surrounding mucosal structures. Therefore, the hybrid encoder allows the network to retain texture sensitivity in early stages while acquiring stronger semantic discrimination in later stages.

To alleviate the loss of boundary-sensitive details at the deepest stage, the bottleneck feature x_4 is further processed by a wavelet bottleneck, as shown in Fig. 2(e). A Haar discrete wavelet transform (DWT) is first applied to decompose x_4 into one low-frequency component LL and three high-frequency components LH, HL and HH. The low-frequency branch preserves coarse semantic structure and is compressed by a lightweight 1×1 projection, while the concatenated high-frequency branch is sent to a dedicated $HighFrequencyRefiner$. The refined frequency components are then fused through an inverse discrete wavelet transform (IDWT), followed by a 3×3 convolutional fusion layer. This design explicitly compensates for the common tendency of deep encoder-decoder networks to lose high-frequency lesion details during semantic compression.

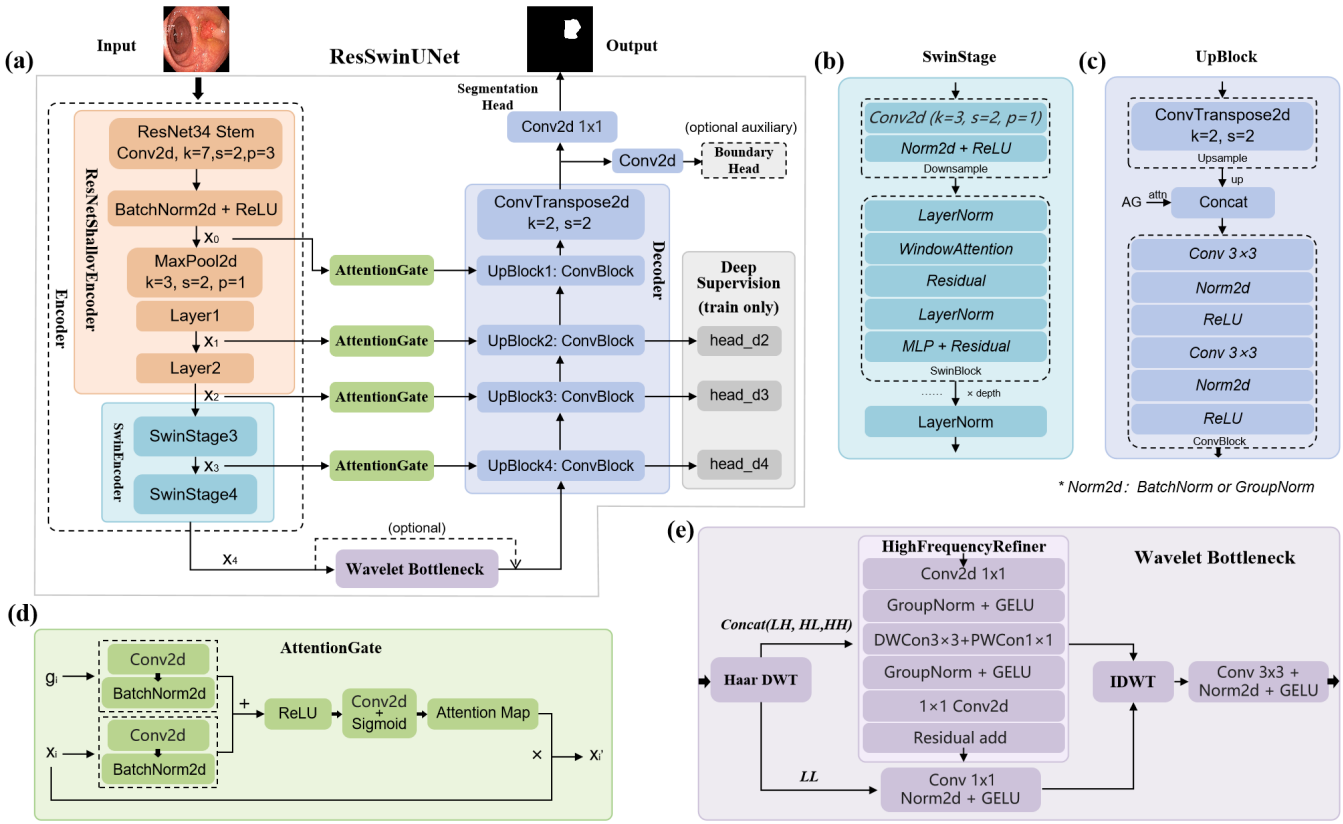

Fig. 2. Overview of the proposed frequency-decoupled ResSwinUNet student network. (a) Overall architecture of the encoder-decoder framework, consisting of a shallow ResNet encoder, Swin-based semantic stages, a wavelet bottleneck, an attention-gated decoder, and prediction heads. (b) Internal structure of the Swin stage, including convolutional downsampling and repeated Swinblocks. (c) Upsampling block used in the decoder for feature fusion and spatial recovery. (d) Attention gate applied on skip connections to suppress irrelevant shallow responses. (e) Wavelet bottleneck, where Haar DWT decomposes the bottleneck feature into low- and high-frequency components, followed by dedicated high-frequency refinement and IDWT-based reconstruction.

The decoder follows a U-shaped down-top pathway and progressively restores spatial resolution through a series of upsampling blocks, as illustrated in Fig. 2(c). Each upsampling block concatenates the decoder feature with the corresponding encoder skip feature,and refines the fused representation.To suppress irrelevant shallow activations before skip fusion, an $AttentionGate$ is inserted on each skip pathway, as shown in Fig. 2(d).The decoder-side gating signal g_i and the encoder-side skip feature x_i are activated and converted into an attention map, which is used to modulate the skip feature element-wise.By filtering out background noise and redundant mucosal textures, the attention gate improves the quality of multi-scale fusion and reduces the risk of propagating distracting shallow patterns into the decoder.

Starting from the refined bottleneck representation, the decoder produces hierarchical features $\{d_4, d_3, d_2, d_1\}$, where intermediate decoder outputs may optionally be equipped with auxiliary supervision heads for training stabilization. The final full-resolution feature is mapped to the segmentation logits by a lightweight prediction head. In our final framework, contour sensitivity is primarily attributed to the wavelet- bottleneck and the logits-edge consistency supervision, while auxiliary heads, if used, are treated as optional components rather than the core contribution.

F. Edge Consistency and Distillation Learning

Instead of relying solely on a heavy explicit boundary branch, we introduce a logits-edge consistency constraint to regularize the main segmentation output. Specifically, edge responses are extracted from the predicted segmentation logits using a gradient operator such as Sobel filtering, and are then matched against boundary maps derived from the corresponding masks. This strategy enables the network to learn contour-aware representations directly from the main segmentation branch, thereby enhancing boundary sharpness with limited structural overhead.

In addition to pseudo-label supervision, a teacher-student distillation scheme is adopted to transfer soft structural priors from the adapted teacher to the student. The first component is response distillation, which encourages the student's probability map to align with the teacher's soft mask output rather than only its binarized pseudo label. The second component is edge distillation, which transfers contour-related knowledge from the teacher to the student. Together, these two forms of distillation help the student learn not only hard region assignments but also smoother uncertainty-aware structural cues.

G. Joint Optimization Strategy

The overall training objective of the student model is formulated as:

$$L_{\text{total}} = \lambda_1 L_{\text{sup}} + \lambda_2 L_{\text{pseudo}} + \lambda_3 L_{\text{edge}} + \lambda_4 L_{\text{distill}} \quad (1)$$

Here, L_{sup} denotes the supervised segmentation loss on L_{small}:

$$L_{\text{sup}} = \frac{1}{|S_{\text{sup}}|} \sum_{i \in S_{\text{sup}}} (w_f L_{\text{focal}}^{(i)} + w_d L_{\text{dice}}^{(i)})$$

where $S_{\text{sup}} = \{i \mid is_labeled = 1, is_pseudo = 0\}$;

L_{pseudo} denotes the confidence-weighted pseudo-label loss on U_{large},:

$$L_{\text{pseudo}} = \frac{\sum_{i \in S_{\text{pseudo}}} \omega_i L_{\text{seg}}^{(i)}}{\sum_{i \in S_{\text{pseudo}}} \omega_i}$$

where $S_{\text{pseudo}} = \{i \mid is_pseudo = 1\}$;

L_{edge} denotes the edge consistency loss:

$$L_{\text{edge}} = \frac{\sum_{i \in S_{\text{edge}}} \omega_i^{edge} L_{\text{edge}}^{(i)}}{\sum_{i \in S_{\text{edge}}} \omega_i^{edge}}$$

and L_{distill} denotes the teacher-student distillation loss:

$$L_{\text{distill}} = \underbrace{\text{mean}_i \ \text{mean}_{pix}(-q_i \log s_i - (1 - q_i) \log(1 - s_i))}_{L_{\text{soft}}} + \underbrace{\text{mean}_i \ \text{mean}_{pix} |\text{Sobel}(\sigma(logits_i)) - distill_edge_i|}_{L_{\text{edge-distill}}}$$

where $s_i = \sigma(logits_i/T), q_i = distill_soft_i$.

In practice: L_{sup} can be implemented as a combination of Dice loss and Focal loss or BCE-based loss to balance region overlap and hard-sample learning.

To mitigate optimization conflict among these objectives, we adopt a joint strategy composed of structural decoupling, spatial task allocation, stage-wise training, and dynamic loss weighting. Structural decoupling is achieved by the wavelet-based separation of low-frequency semantics and high-frequency boundaries, reducing direct competition within a single shared representation. Spatial task allocation assigns different supervision signals to different regions, where high-confidence regions receive harder pseudo-label supervision, uncertain boundary bands receive softer distillation, and narrow ground-truth boundary regions are emphasized by the edge-consistency term. Stage-wise training progressively introduces losses, typically starting from real-label segmentation, then adding edge consistency, followed by pseudo-label supervision, and finally distillation. Dynamic weighting further ensures that edge loss starts with a small coefficient, pseudo-label loss is gradually warmed up, and distillation is introduced later with a relatively conservative weight.

Through this design, the student is expected to learn robust region prediction, sharper contour modeling, and teacher-guided structural priors without suffering from severe instability caused by naively optimizing all losses with equal strength from the beginning. The final output of the framework is therefore a frequency-decoupled, edge-sensitive, and deployment-efficient ResSwinUNet student model for automatic colonoscopic polyp segmentation under low-annotation-budget conditions.

IV. RESULTS

A. Evaluation Metrics

See [1]–[5] for resources on formatting math into text and additional help in working with LaTeX.

B.

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D.

V. TEXT

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VI. Some Common Elements

A. Sections and Subsections

Enumeration of section headings is desirable, but not required. When numbered, please be consistent throughout the article, that is, all headings and all levels of section headings in the article should be enumerated. Primary headings are designated with Roman numerals, secondary with capital letters, tertiary with Arabic numbers; and quaternary with lowercase letters. Reference and Acknowledgment headings are unlike all other section headings in text. They are never enumerated. They are simply primary headings without labels, regardless of whether the other headings in the article are enumerated.

B. Citations to the Bibliography

The coding for the citations is made with the LaTeX `\cite` command. This will display as: see [1].

For multiple citations code as follows: `\cite{ref1,ref2,ref3}` which will produce [1]–[3]. For reference ranges that are not consecutive code as `\cite{ref1,ref2,ref3,ref9}` which will produce [1]–[3], [9]

C. Lists

In this section, we will consider three types of lists: simple unnumbered, numbered, and bulleted. There have been many options added to IEEEtran to enhance the creation of lists. If your lists are more complex than those shown below, please refer to the original "IEEEtran_HOWTO.pdf" for additional options.

A plain unnumbered list:

bare_jrnl.tex
bare_conf.tex
bare_jrnl_compsoc.tex
bare_conf_compsoc.tex
bare_jrnl_comsoc.tex

Fig. 3. Simulation results for the network.

A simple numbered list:

1) bare_jrnl.tex
2) bare_conf.tex
3) bare_jrnl_compsoc.tex
4) bare_conf_compsoc.tex
5) bare_jrnl_comsoc.tex

A simple bulleted list:

- bare_jrnl.tex
- bare_conf.tex
- bare_jrnl_compsoc.tex
- bare_conf_compsoc.tex
- bare_jrnl_comsoc.tex

D. Figures

Fig. 1 is an example of a floating figure using the graphicx package. Note that `\label` must occur AFTER (or within) `\caption`. For figures, `\caption` should occur after the `\includegraphics`.

Fig. 2(a) and 2(b) is an example of a double column floating figure using two subfigures. (The subfig.sty package must be loaded for this to work.) The subfigure `\label` commands are set within each subfloat command, and the `\label` for the overall figure must come after `\caption`. `\hfil` is used as a separator to get equal spacing. The combined width of all the parts of the figure should do not exceed the text width or a line break will occur.

Note that often IEEE papers with multi-part figures do not place the labels within the image itself (using the optional argument to `\subfloat[]`), but instead will reference/describe all of them (a), (b), etc., within the main caption. Be aware that for subfig.sty to generate the (a), (b), etc., subfigure labels, the optional argument to `\subfloat` must be present. If a subcaption is not desired, leave its contents blank, e.g.,`\subfloat[]`.

VII. Tables

Note that, for IEEE-style tables, the `\caption` command should come BEFORE the table. Table captions use title case. Articles (a, an, the), coordinating conjunctions (and, but, for,

(a) (b)

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TABLE II
AN EXAMPLE OF A TABLE

One	Two
Three	Four

or, nor), and most short prepositions are lowercase unless they are the first or last word. Table text will default to `\footnotesize` as the IEEE normally uses this smaller font for tables. The `\label` must come after `\caption` as always.

VIII. ALGORITHMS

Algorithms should be numbered and include a short title. They are set off from the text with rules above and below the title and after the last line.

Algorithm 1 Weighted Tanimoto ELM.

TRAIN($\mathbf{XT}$)
  select randomly $W \subset \mathbf{X}$
  $N_{\mathbf{t}} \leftarrow |\{i : \mathbf{t}_i = \mathbf{t}\}|$ **for** $\mathbf{t} = -1, +1$
  $B_i \leftarrow \sqrt{\text{MAX}(N_{-1}, N_{+1})/N_{\mathbf{t}_i}}$ **for** $i = 1, ..., N$
  $\hat{\mathbf{H}} \leftarrow B \cdot (\mathbf{X}^T\mathbf{W})/(\not{\mathbb{1}}\mathbf{X} + \not{\mathbb{1}}\mathbf{W} - \mathbf{X}^T\mathbf{W})$
  $\beta \leftarrow \left(I/C + \hat{\mathbf{H}}^T\hat{\mathbf{H}}\right)^{-1} (\hat{\mathbf{H}}^T B \cdot \mathbf{T})$
  return $\mathbf{W}, \beta$

PREDICT($\mathbf{X}$)
  $\mathbf{H} \leftarrow (\mathbf{X}^T\mathbf{W})/(\not{\mathbb{1}}\mathbf{X} + \not{\mathbb{1}}\mathbf{W} - \mathbf{X}^T\mathbf{W})$
  return SIGN($\mathbf{H}\beta$)

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IX. MATHEMATICAL TYPOGRAPHY AND WHY IT MATTERS

Typographical conventions for mathematical formulas have been developed to **provide uniformity and clarity of presentation across mathematical texts**. This enables the readers of those texts to both understand the author's ideas and to grasp new concepts quickly. While software such as LaTeX and MathType® can produce aesthetically pleasing math when used properly, it is also very easy to misuse the software, potentially resulting in incorrect math display.

IEEE aims to provide authors with the proper guidance on mathematical typesetting style and assist them in writing the best possible article. As such, IEEE has assembled a set of examples of good and bad mathematical typesetting [1]–[5].

Further examples can be found at http://journals.ieeeauthorcenter.ieee.org/wp-content/uploads/sites/7/IEEE-Math-Typesetting-Guide-for-LaTeX-Users.pdf

A. Display Equations

The simple display equation example shown below uses the "equation" environment. To number the equations, use the `\label` macro to create an identifier for the equation. LaTeX will automatically number the equation for you.

$$x = \sum_{i=0}^{n} 2iQ. \tag{3}$$

is coded as follows:

```
\begin{equation}
\label{deqn_ex1}
x = \sum_{i=0}^{n} 2{i} Q.
\end{equation}
```

To reference this equation in the text use the `\ref` macro. Please see (3)
is coded as follows:

```
Please see (\ref{deqn_ex1})
```

B. Equation Numbering

Consecutive Numbering: Equations within an article are numbered consecutively from the beginning of the article to the end, i.e., (1), (2), (3), (4), (5), etc. Do not use roman numerals or section numbers for equation numbering.
Appendix Equations: The continuation of consecutively numbered equations is best in the Appendix, but numbering as (A1), (A2), etc., is permissible.

Hyphens and Periods: Hyphens and periods should not be used in equation numbers, i.e., use (1a) rather than (1-a) and (2a) rather than (2.a) for subequations. This should be consistent throughout the article.

C. Multi-Line Equations and Alignment

Here we show several examples of multi-line equations and proper alignments.
A single equation that must break over multiple lines due to length with no specific alignment.

$$\begin{multline}\text{The first line of this example}\\ \text{The second line of this example}\\ \text{The third line of this example}\end{multline} \tag{4}$$

is coded as:

```
\begin{multline}
\text{The first line of this example}\\
\text{The second line of this example}\\
\text{The third line of this example}
\end{multline}
```

A single equation with multiple lines aligned at the = signs

$$\begin{aligned} a &= c+d \\ b &= e+f \end{aligned} \tag{5, 6}$$

is coded as:

```
\begin{align}
a &= c+d \\
b &= e+f
\end{align}
```

The `align` environment can align on multiple points as shown in the following example:

$$\begin{aligned} x &= y & X &= Y & a &= bc \\ x' &= y' & X' &= Y' & a' &= bz \end{aligned} \tag{7, 8}$$

is coded as:

```
\begin{align}
x &= y & X & =Y & a &=bc\\
x' &= y' & X' &=Y' &a' &=bz
\end{align}
```

D. Subnumbering

The amsmath package provides a `subequations` environment to facilitate subnumbering. An example:

$$\begin{subequations}\label{eq:2}\begin{aligned} f&=g \label{eq:2A}\\ f' &=g' \label{eq:2B}\\ \mathcal{L}f &= \mathcal{L}g \label{eq:2c}\end{aligned}\end{subequations}$$

(9a) $f = g$
(9b) $f' = g'$
(9c) $\mathcal{L}f = \mathcal{L}g$

is coded as:

```
\begin{subequations}\label{eq:2}
\begin{align}
f&=g \label{eq:2A}\\
f' &=g' \label{eq:2B}\\
\mathcal{L}f &= \mathcal{L}g \label{eq:2c}
\end{align}
\end{subequations}
```

E. Matrices

There are several useful matrix environments that can save you some keystrokes. See the example coding below and the output.
A simple matrix:

$$\begin{matrix} 0 & 1 \\ 1 & 0 \end{matrix} \tag{10}$$

is coded as:

```
\begin{equation}
\begin{matrix}  0 &  1 \\
1 &  0 \end{matrix}
\end{equation}
```

A matrix with parenthesis

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \tag{11}$$

is coded as:

```
\begin{equation}
\begin{pmatrix} 0 & -i \\
 i &  0 \end{pmatrix}
\end{equation}
```

A matrix with square brackets

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \tag{12}$$

is coded as:

```
\begin{equation}
\begin{bmatrix} 0 & -1 \\
1 &  0 \end{bmatrix}
\end{equation}
```

A matrix with curly braces

$$\begin{Bmatrix} 1 & 0 \\ 0 & -1 \end{Bmatrix} \tag{13}$$

is coded as:

```
\begin{equation}
\begin{Bmatrix} 1 &  0 \\
0 & -1 \end{Bmatrix}
\end{equation}
```

A matrix with single verticals

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \tag{14}$$

is coded as:

```
\begin{equation}
\begin{vmatrix} a &  b \\
c &  d \end{vmatrix}
\end{equation}
```

A matrix with double verticals

$$\begin{Vmatrix} i & 0 \\ 0 & -i \end{Vmatrix} \tag{15}$$

is coded as:

```
\begin{equation}
\begin{Vmatrix} i &  0 \\
0 & -i \end{Vmatrix}
\end{equation}
```

F. Arrays

The `array` environment allows you some options for matrix-like equations. You will have to manually key the fences, but there are other options for alignment of the columns and for setting horizontal and vertical rules. The argument to `array` controls alignment and placement of vertical rules.

A simple array

$$\left(\begin{array}{cccc} a+b+c & uv & x-y & 27 \\ a+b & u+v & z & 134 \end{array}\right) \tag{16}$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{cccc}
a+b+c & uv & x-y & 27\\
a+b & u+v & z & 134
\end{array} \right)
\end{equation}
```

A slight variation on this to better align the numbers in the last column

$$\left(\begin{array}{cccr} a+b+c & uv & x-y & 27 \\ a+b & u+v & z & 134 \end{array}\right) \tag{17}$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{cccr}
a+b+c & uv & x-y & 27\\
a+b & u+v & z & 134
\end{array} \right)
\end{equation}
```

An array with vertical and horizontal rules

$$\left(\begin{array}{c|c|c|r} a+b+c & uv & x-y & 27 \\ \hline a+b & u+v & z & 134 \end{array}\right) \tag{18}$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{c|c|c|r}
a+b+c & uv & x-y & 27\\
a+b & u+v & z & 134
\end{array} \right)
\end{equation}
```

Note the argument now has the pipe "|" included to indicate the placement of the vertical rules.

G. Cases Structures

Many times cases can be miscoded using the wrong environment, i.e., `array`. Using the `cases` environment will save keystrokes (from not having to type the `\left\lbrace`) and automatically provide the correct column alignment.

$$z_m(t) = \begin{cases} 1, & \text{if } \beta_m(t) \\ 0, & \text{otherwise.} \end{cases}$$

is coded as follows:

```
\begin{equation*}
{z_m(t)} =
\begin{cases}
1,&{\text{if}}\ {\beta }_m(t),\\
{0,}&{\text{otherwise.}}
\end{cases}
\end{equation*}
```

Note that the "&" is used to mark the tabular alignment. This is important to get proper column alignment. Do not use `\quad` or other fixed spaces to try and align the columns. Also, note the use of the `\text` macro for text elements such as "if" and "otherwise."

H. Function Formatting in Equations

Often, there is an easy way to properly format most common functions. Use of the \ in front of the function name will in most cases, provide the correct formatting. When this does not work, the following example provides a solution using the `\text` macro:

$$d_R^{KM} = \underset{d_t^{KM}}{\text{arg min}} \{ d_1^{KM}, \ldots, d_6^{KM}\}.$$

is coded as follows:

```
\begin{equation*}
 d_{R}^{KM} = \underset {d_{l}^{KM}}
 {\text{arg min}} \{ d_{1}^{KM},
 \ldots,d_{6}^{KM}\}.
\end{equation*}
```

I. Text Acronyms Inside Equations

This example shows where the acronym "MSE" is coded using `\text{}` to match how it appears in the text.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y_i})^2$$

```
\begin{equation*}
 \text{MSE} = \frac {1}{n}\sum _{i=1}^{n}
(Y_{i} - \hat {Y_{i}})^{2}
\end{equation*}
```

X. Conclusion

The conclusion goes here.

Acknowledgments

This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this article.

Appendix
Proof of the Zonklar Equations

Use `\appendix` if you have a single appendix: Do not use `\section` anymore after `\appendix`, only `\section*`. If you have multiple appendixes use `\appendices` then use `\section` to start each appendix. You must declare a `\section` before using any `\subsection` or using `\label` (`\appendices` by itself starts a section numbered zero.)

References Section

You can use a bibliography generated by BibTeX as a .bbl file. BibTeX documentation can be easily obtained at: http://mirror.ctan.org/biblio/bibtex/contrib/doc/ The IEEEtran BibTeX style support page is: http://www.michaelshell.org/tex/ieeetran/bibtex/

Simple References

You can manually copy in the resultant .bbl file and set second argument of `\begin` to the number of references (used to reserve space for the reference number labels box).

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Biography Section

If you have an EPS/PDF photo (graphicx package needed), extra braces are needed around the contents of the optional argument to biography to prevent the LaTeX parser from getting confused when it sees the complicated `\includegraphics` command within an optional argument. (You can create your own custom macro containing the `\includegraphics` command to make things simpler here.)

If you include a photo:

Michael Shell Use `\begin{IEEEbiography}` and then for the 1st argument use `\includegraphics` to declare and link the author photo. Use the author name as the 3rd argument followed by the biography text.

If you will not include a photo:

John Doe Use `\begin{IEEEbiographynophoto}` and the author name as the argument followed by the biography text.